Problem 1

Let $V = \mathbb{R}^n$ and M be an $n \times n$ real matrix. This exercise shows that $\|u\| = (u^* M u)^{1/2}$ is a vector norm whenever M is *positive definite* (defined below).

(a) Show that $u^*Mu = u^*M^*u = u^*\left(\frac{1}{2}(M + M^*)\right)u$ for all $u \in V$. This means that as long as we consider functions of the form $f(u) = u^*Mu$, we may assume M is symmetric. For the rest of this question, assume M is symmetric. Hint: u^*Mu is a 1×1 matrix and therefore equal to its transpose.

(b) Show that the function $\|u\| = (u^*Mu)^{1/2}$ is homogeneous: $\|au\| = |a|\,\|u\|$.

(c) We say M is positive definite if $u^*Mu > 0$ whenever $u \neq 0$. Show that if M is positive definite, then $\|u\| \geq 0$ for all u and $\|u\| = 0$ only for $u = 0$.

(d) Show that if M is symmetric and positive definite (SPD), then $|u^*Mv| \leq \|u\|\,\|v\|$. This is the *Cauchy–Schwarz inequality*. Hint (a famous old trick): $\phi(t) = (u + tv)^*M(u + tv)$ is a quadratic function of t that is non-negative for all t if M is positive definite. The Cauchy–Schwarz inequality follows from requiring that the minimum value of ϕ is not negative, assuming $M^* = M$.

(e) Use the Cauchy–Schwarz inequality to verify the triangle inequality in its squared form $\|u + v\|^2 \leq \|u\|^2 + 2\,\|u\|\,\|v\| + \|v\|^2$.

(f) Show that if $M = I$ then $\|u\|$ is the l^2 norm of u.

Solution.

(a) Since u^*Mu is a 1×1 matrix, it equals its own transpose:

$$u^*Mu = (u^*Mu)^T = u^TM^Tu = u^*M^*u.$$

Adding this to the identity $u^*Mu = u^*Mu$ gives $2\,u^*Mu = u^*(M + M^*)u$, hence

$$u^*Mu = u^*\left(\tfrac{1}{2}(M + M^*)\right)u.$$

Since $\frac{1}{2}(M + M^*)$ is symmetric, any study of the quadratic form $f(u) = u^*Mu$ may assume M is symmetric without loss of generality.

(b) For any scalar $a \in \mathbb{R}$,

$$\|au\| = \big((au)^*M(au)\big)^{1/2} = \big(a^2\,u^*Mu\big)^{1/2} = |a|\,(u^*Mu)^{1/2} = |a|\,\|u\|.$$

(c) If M is positive definite, then $u^*Mu > 0$ for all $u \neq 0$, so $\|u\| = (u^*Mu)^{1/2} > 0$. For $u = 0$, we have $\|0\| = (0^TM \cdot 0)^{1/2} = 0$. Therefore $\|u\| \geq 0$ for all u, and $\|u\| = 0$ if and only if $u = 0$.

(d) Define $\phi(t) = (u+tv)^*M(u+tv)$ for $t \in \mathbb{R}$. Since M is positive definite, $\phi(t) \geq 0$ for all t. Expanding, and using $M = M^*$ so that $v^*Mu = (u^*Mv)^T = u^*Mv$:

$$\phi(t) = u^*Mu + 2t\,(u^*Mv) + t^2\, v^*Mv = \|v\|^2 t^2 + 2(u^*Mv)\, t + \|u\|^2.$$

This is a quadratic in t with non-negative minimum, so the discriminant must satisfy

$$4(u^*Mv)^2 - 4\,\|u\|^2\,\|v\|^2 \leq 0 \quad \Longrightarrow \quad |u^*Mv| \leq \|u\|\,\|v\|.$$

(e) Expanding $\|u+v\|^2$ and using $M = M^*$:

$$\|u+v\|^2 = (u+v)^*M(u+v) = u^*Mu + 2\,u^*Mv + v^*Mv = \|u\|^2 + 2\,u^*Mv + \|v\|^2.$$

Applying the Cauchy–Schwarz inequality from part (d), $u^*Mv \leq |u^*Mv| \leq \|u\|\,\|v\|$, so

$$\|u+v\|^2 \leq \|u\|^2 + 2\,\|u\|\,\|v\| + \|v\|^2 = \big(\|u\| + \|v\|\big)^2.$$

(f) If $M = I$, then $u^*Mu = u^*u = \sum_{i=1}^n u_i^2$, so

$$\|u\| = \left(\sum_{i=1}^n u_i^2\right)^{1/2},$$

which is the l^2 norm of u.

Problem 2

Suppose that A is an $n \times n$ invertible matrix. Show that

$$\|A^{-1}\| = \max_{u \neq 0} \frac{\|u\|}{\|Au\|} = \left(\min_{u \neq 0} \frac{\|Au\|}{\|u\|}\right)^{-1}.$$

Solution.

By definition of the operator norm,

$$\|A^{-1}\| = \max_{v \neq 0} \frac{\|A^{-1}v\|}{\|v\|}.$$

Since A is invertible, the map $u \mapsto Au$ is a bijection on $\mathbb{R}^n \setminus \{0\}$. Substituting $v = Au$ (so $A^{-1}v = u$), as v ranges over all nonzero vectors, so does u:

$$\|A^{-1}\| = \max_{u \neq 0} \frac{\|u\|}{\|Au\|}.$$

For the second equality, note that

$$\max_{u \neq 0} \frac{\|u\|}{\|Au\|} = \max_{u \neq 0} \left(\frac{\|Au\|}{\|u\|}\right)^{-1} = \left(\min_{u \neq 0} \frac{\|Au\|}{\|u\|}\right)^{-1},$$

where the last step uses the fact that the maximum of $1/g(u)$ equals the reciprocal of the minimum of $g(u)$, valid since $\|Au\|/\|u\| > 0$ for all $u \neq 0$ (by invertibility of A).

Problem 3

The *symmetric part* of the real $n \times n$ matrix is $M = \frac{1}{2}(A + A^*)$. Show that $\nabla\bigl(\frac{1}{2}x^*Ax\bigr) = Mx$.

Solution.

Define $f(x) = \frac{1}{2}x^*Ax$. This is $\frac{1}{2}$ times a product of the $1 \times n$ matrix x^* with the $n \times n$ matrix A with the $n \times 1$ matrix x. By the Leibniz (product) rule:

$$\dot{f} = \tfrac{1}{2}\bigl(\dot{x}^*Ax + x^*A\dot{x}\bigr).$$

Since $\dot{x}^*Ax$ is a 1×1 real matrix, it equals its own transpose:

$$\dot{x}^*Ax = (\dot{x}^*Ax)^T = x^TA^T\dot{x} = x^*A^*\dot{x}.$$

Therefore

$$\dot{f} = \tfrac{1}{2}\bigl(x^*A^*\dot{x} + x^*A\dot{x}\bigr) = \tfrac{1}{2}\,x^*(A + A^*)\dot{x}.$$

Since ∇f is defined by the relation $\dot{f} = (\nabla f)^*\dot{x}$ for all $\dot{x}$, we read off:

$$\nabla f = \tfrac{1}{2}(A + A^*)x = Mx.$$

Problem 4

The *informal* condition number of the problem of computing the action of A is

$$\kappa(A) = \max_{x \neq 0,\, \Delta x \neq 0} \frac{\|A(x + \Delta x) - Ax\|/\|Ax\|}{\|x + \Delta x\|/\|x\|}.$$

Alternatively, it is the sharp constant in the estimate

$$\frac{\|A(x + \Delta x) - Ax\|}{\|Ax\|} \leq C \cdot \frac{\|x + \Delta x\|}{\|x\|},$$

which bounds the worst case relative change in the answer in terms of the relative change in the data. Show that for the l^2 norm,

$$\kappa = \sigma_{\max}/\sigma_{\min},$$

the ratio of the largest to smallest singular value of A. Show that this formula holds even when A is not square.

Solution.

Since A is linear, $A(x + \Delta x) - Ax = A\Delta x$. Maximizing over Δx for a fixed x, we may replace Δx by any nonzero vector independently of x. Writing the ratio as a product:

$$\kappa(A) = \max_{\Delta x \neq 0} \frac{\|A\Delta x\|}{\|\Delta x\|} \cdot \max_{x \neq 0} \frac{\|x\|}{\|Ax\|} = \left(\max_{x \neq 0} \frac{\|Ax\|}{\|x\|}\right) \cdot \left(\min_{x \neq 0} \frac{\|Ax\|}{\|x\|}\right)^{-1}.$$

Now let A be $m \times n$ (not necessarily square) with full column rank, and let $A = U\Sigma V^*$ be its SVD, where U is $m \times m$ orthogonal, V is $n \times n$ orthogonal, and Σ is $m \times n$ with diagonal entries $\sigma_1 \geq \cdots \geq \sigma_n > 0$. Substituting $y = V^* x$ (a bijection on $\mathbb{R}^n$ since V is orthogonal), and using $\|Ux\|_2 = \|x\|_2$ for any orthogonal U:

$$\frac{\|Ax\|_2}{\|x\|_2} = \frac{\|U\Sigma V^* x\|_2}{\|x\|_2} = \frac{\|\Sigma y\|_2}{\|y\|_2} = \frac{\left(\sum_{i=1}^n \sigma_i^2\, y_i^2\right)^{1/2}}{\left(\sum_{i=1}^n y_i^2\right)^{1/2}}.$$

This is a weighted RMS average of the σ_i, which is maximized when all weight is on σ_1 (i.e. $y = e_1$) and minimized when all weight is on σ_n (i.e. $y = e_n$). Therefore

$$\max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_{\max}, \qquad \min_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_{\min},$$

and so $\kappa(A) = \sigma_{\max}/\sigma_{\min}$. This derivation uses only V^* being a bijection on $\mathbb{R}^n$ and U preserving the l^2 norm, both of which hold regardless of whether A is square, so the formula is valid for non-square A as well.

Problem 5

Show that a symmetric $n \times n$ real matrix is positive definite if and only if all its eigenvalues are positive. Hint: If R is a right eigenvector matrix, then, for a symmetric matrix we may normalize so that $R^{-1} = R^*$ and $A = R\Lambda R^*$, where Λ is a diagonal matrix containing the eigenvalues (Sections 4.2.5 and 4.2.7). Then $x^* A x = x^* R \Lambda R^* x = (x^* R)\Lambda(R^* x) = y^* \Lambda y$, where $y = R^* x$. Show that $y^* \Lambda y > 0$ for all $y \neq 0$ if and only if all the diagonal entries of Λ are positive. Show that if A is positive definite, then there is a $C > 0$ so that $x^* A x > C \|x\|_{l^2}^2$ for all x. (Hint: $\|x\|_{l^2} = \|x\|_{l^2}$, $C = \lambda_{\min}$.)

Solution.

Since A is real symmetric, the spectral theorem guarantees an orthogonal eigenvector matrix R with $R^{-1} = R^*$, so that $A = R\Lambda R^*$ with $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. For any $x \in \mathbb{R}^n$, set $y = R^* x$. Since R is orthogonal, $y = 0$ if and only if $x = 0$, and $\|y\|_{l^2} = \|x\|_{l^2}$. Then:

$$x^* A x = x^* R \Lambda R^* x = y^* \Lambda y = \sum_{i=1}^n \lambda_i\, y_i^2.$$

($\Leftarrow$) If all $\lambda_i > 0$ and $x \neq 0$, then $y \neq 0$, so at least one $y_i \neq 0$. Every term $\lambda_i y_i^2 \geq 0$ and at least one is strictly positive, so $x^* A x > 0$. Hence A is positive definite.

($\Rightarrow$) If A is positive definite, then for each eigenvector r_k (the k-th column of R), we have $0 < r_k^* A r_k = \lambda_k\, r_k^* r_k = \lambda_k$. So all eigenvalues are positive.

For the lower bound, since each $\lambda_i \geq \lambda_{\min} > 0$:

$$x^* A x = \sum_{i=1}^n \lambda_i\, y_i^2 \geq \lambda_{\min} \sum_{i=1}^n y_i^2 = \lambda_{\min}\, \|y\|_{l^2}^2 = \lambda_{\min}\, \|x\|_{l^2}^2.$$

Thus $x^*Ax \geq C \|x\|_{l^2}^2$ for all x with $C = \lambda_{\min} > 0$.

Problem 6

Suppose $v_1, \ldots, v_m$ is an orthonormal basis for a vector space $V \subseteq \mathbb{R}^n$. Let L be a linear transformation from V to V. Let A be the matrix that represents L in this basis. Show that the entries of A are given by

$$a_{jk} = v_j^* L v_k. \tag{5.8}$$

Hint: Show that if $y \in V$, the representation of y in this basis is $y = \sum_j y_j v_j$, where $y_j = v_j^* y$. In physics and theoretical chemistry, inner products of the form (5.8) are called *matrix elements*. For example, the eigenvalue perturbation formula (4.39) (in physicist terminology) simply says that the perturbation in an eigenvalue is (nearly) equal to the appropriate matrix element of the perturbation of the matrix.

Solution.

Since $\{v_1, \ldots, v_m\}$ is an orthonormal basis for V, any $y \in V$ can be written as $y = \sum_{j=1}^m y_j v_j$. To find the coefficient y_j, take the inner product with v_j:

$$v_j^* y = v_j^* \sum_{i=1}^m y_i v_i = \sum_{i=1}^m y_i \, v_j^* v_i = \sum_{i=1}^m y_i \, \delta_{ji} = y_j,$$

using orthonormality $v_j^* v_i = \delta_{ji}$.

Now, A represents L in this basis, meaning that if $y = \sum_k y_k v_k$, then $Ly = \sum_j (Ay)_j \, v_j$ where $(Ay)_j = \sum_k a_{jk} \, y_k$. Apply L to a basis vector v_k: since v_k has representation $y_i = \delta_{ik}$, we get

$$Lv_k = \sum_{j=1}^m a_{jk} \, v_j.$$

Taking the inner product of both sides with v_j:

$$v_j^* L v_k = v_j^* \sum_{i=1}^m a_{ik} \, v_i = \sum_{i=1}^m a_{ik} \, \delta_{ji} = a_{jk}.$$

Problem 7

Suppose A is an $n \times n$ symmetric matrix and $V \subset \mathbb{R}^n$ is an *invariant subspace* for A (i.e. $Ax \in V$ if $x \in V$). Show that A defines a linear transformation from V to V. Show that there is a basis for V in which this linear transformation (called A *restricted to* V) is represented by a symmetric matrix. Hint: construct an orthonormal basis for V.

Solution.

A defines a linear transformation from V to V: Let $L : V \to V$ be defined by $L(x) = Ax$. Since V is an invariant subspace, $Ax \in V$ for all $x \in V$, so L is well-defined. Linearity follows from A being a matrix: $L(\alpha x + \beta y) = A(\alpha x + \beta y) = \alpha Ax + \beta Ay = \alpha L(x) + \beta L(y)$.

The matrix representation is symmetric: Let $v_1, \ldots, v_m$ be an orthonormal basis for V (obtained, for example, by Gram–Schmidt). By the result of Problem 6 (eq. 5.8), the matrix B representing L in this basis has entries

$$b_{jk} = v_j^* A v_k.$$

Since A is symmetric ($A = A^*$), we have

$$b_{kj} = v_k^* A v_j = v_k^* A^* v_j = (Av_k)^* v_j = (v_j^* A v_k)^T = v_j^* A v_k = b_{jk},$$

where the second-to-last equality uses the fact that $v_j^* A v_k$ is a 1×1 matrix and hence equals its own transpose. Therefore $B = B^*$, i.e., B is symmetric.

Problem 8

If Q is an $n \times n$ matrix, and $(Qx)^* Qy = x^* y$ for all x and y, show that Q is an orthogonal matrix. Hint: If $(Qx)^* Qy = x^*(Q^*Q)y = x^* y$, we can explore the entries of Q^*Q by choosing particular vectors x and y.

Solution.

The hypothesis $(Qx)^* Qy = x^* y$ for all x, y can be rewritten as

$$x^* (Q^*Q)\, y = x^* y \quad \text{for all } x, y \in \mathbb{R}^n.$$

This means $x^*(Q^*Q - I)\, y = 0$ for all x and y. Let $M = Q^*Q - I$. Choose $x = e_j$ and $y = e_k$ (the standard basis vectors):

$$e_j^*\, M\, e_k = M_{jk} = 0 \quad \text{for all } j, k.$$

Since every entry of M is zero, $M = 0$, i.e. $Q^*Q = I$. This is the definition of an orthogonal matrix.

Problem 9

If $\|Qx\|_{l^2} = \|x\|_{l^2}$ for all x, show that $(Qx)^* Qy = x^* y$ for all x and y. Hint (*polarization*): If $\|Q(x + sy)\|_{l^2}^2 = \|x + sy\|_{l^2}^2$ for all s, then $(Qx)^* Qy = x^* y$.

Solution.

Since $\|Qx\|_{l^2} = \|x\|_{l^2}$ for all x, in particular it holds for x replaced by $x + sy$ for any scalar s:

$$\|Q(x + sy)\|_{l^2}^2 = \|x + sy\|_{l^2}^2.$$

Expand the left side using linearity of Q:

$$\|Q(x + sy)\|_{l^2}^2 = (Qx + sQy)^*(Qx + sQy) = (Qx)^*Qx + 2s\,(Qx)^*Qy + s^2\,(Qy)^*Qy.$$

Expand the right side:

$$\|x + sy\|_{l^2}^2 = (x + sy)^*(x + sy) = x^*x + 2s\,x^*y + s^2\,y^*y.$$

By hypothesis, $\|Qx\|_{l^2}^2 = \|x\|_{l^2}^2$ and $\|Qy\|_{l^2}^2 = \|y\|_{l^2}^2$, so the s^0 and s^2 terms on both sides are equal. Matching the coefficient of s (the linear term) gives:

$$2\,(Qx)^*Qy = 2\,x^*y,$$

and therefore $(Qx)^*Qy = x^*y$ for all x and y.